Halo Mass Mediates the Colour–Environment Relation in GAMA: Evidence from Partial Correlation and Distribution Shape Analysis

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Abstract

We investigate the relationship between galaxy colour (u−r) and local environment for galaxies in the GAMA DR4 survey, using partial correlation to control for group halo mass and bootstrap null tests to assess physical evolution in the shape of the colour distribution with redshift. For elliptical galaxies (N = 1,270), raw correlations between u−r and all four environment metrics collapse to $r \approx 0$ once group halo mass is controlled, demonstrating complete mediation. For the full galaxy population (N = 22,531), ~50% of the signal survives halo control, indicating an independent morphology-linked pathway. Skewness and Box–Cox λ of the u−r distribution evolve monotonically with redshift (Spearman $|r| = 1.00$, $p < 0.001$ for both): skewness shifts from +0.90 at z = 0.03 to −0.82 at z = 0.33, and λ from −0.20 to +2.70. Both trends exceed the bootstrap 95% null envelope at every redshift bin, confirming the trends are real features of the observed sample. Because GAMA is flux-limited at $r < 19.8$, the trends reflect changing sample composition with redshift (Malmquist selection) as well as any genuine evolution; a mass-complete subsample is required to separate the two. Comparison of λ with the red fraction (u−r > 2.1) shows perfect rank agreement (Spearman r = 1.000) but diverging rates, illustrating that λ captures distributional information — the sharpness and symmetry of the colour distribution — that a fixed threshold cannot.

1. Introduction

The correlation between galaxy colour and local environment is among the most robust observational results in galaxy evolution (Dressler 1980; Blanton & Moustakas 2009). The outstanding question is mechanistic: does environment directly drive quenching, or is the signal primarily a reflection of halo mass, which correlates with both environment and the efficiency of quenching processes such as ram-pressure stripping and strangulation?

A complementary question is whether the shape of the colour distribution encodes information beyond the mean or a simple threshold count. If the u−r distribution transforms systematically with redshift in ways exceeding sampling noise, that transformation may constrain the quenching process. Box–Cox λ characterises the full distributional shape, including the breadth and asymmetry of the red sequence, and can therefore detect changes invisible to a fixed colour threshold such as f_red = fraction above u−r = 2.1. Understanding when the two statistics agree or diverge informs the choice of statistical estimator for future redshift-resolved quenching analyses.

We address both questions using GAMA DR4 (Driver et al. 2022), exploiting its group catalogue (Robotham et al. 2011) and environment measures (Brough et al. 2013). We are explicit throughout about the role of Malmquist selection in a flux-limited sample.

2. Data and Sample

GAMA DR4 StellarMassesLambdarv24 (200,077 galaxies), quality cuts 0.5 < u−r < 4.0, log M_* > 8.0, 0.002 < z < 0.35: 165,488 galaxies for shape analysis. Partial correlation additionally uses G3CGalv10 (group membership), G3CFoFGroupv10 (halo mass proxy MassA), and EnvironmentMeasuresv06 (CountInCyl, DistanceTo5nn, SurfaceDensity, AGEDenPar). Elliptical subsample: VisualMorphologyv03 ELLIPTICAL_CODE = 1, N = 1,270. Full group population N = 22,531. GAMA is flux-limited at r < 19.8; at z $\gtrsim$ 0.2 the survey is increasingly incomplete for low-mass, blue galaxies and the visible population is dominated by massive early-type galaxies. Trends with redshift in the flux-limited sample must therefore be interpreted with this selection in mind.

3. Methods

3.1 Partial Correlation

We residualise both u−r and each log-environment measure against log(MassA) via linear regression and correlate residuals (Pearson and Spearman). Results at individual galaxy level and group level (mean u−r; red fraction f_red = fraction with u−r > 2.1).

3.2 Distribution Shape Evolution

Seven equal-width redshift bins ($z = 0.002–0.35$, $\Delta z = 0.05$; N = 3,900–34,000). In each bin we compute skewness, excess kurtosis, and Box–Cox λ. Johnson SU parameters are reported in Appendix A; they show numerical instability and are excluded from the main analysis. We bootstrap 500 same-sized subsamples from the full pool to build 95% null envelopes and apply Spearman and Kendall rank tests for monotonic trend.

3.3 Comparison of λ and Red Fraction

Red fraction f_red is computed in each redshift bin as the fraction of galaxies with u−r > 2.1. Both f_red and λ are normalised to [0,1] over the seven bins to place them on a common scale and compared as distributional statistics measured on the same flux-limited samples. Because the underlying sample composition changes with redshift, any differences in the rates at which the two statistics change with z reflect both physical effects and the different mathematical responses of the statistics to Malmquist-selected samples. Conclusions about physical processes are therefore limited to the existence and direction of the trends, not their absolute rates.

4. Results

4.1 Halo Mass Fully Mediates Colour–Environment for Ellipticals

Table 1 shows raw and partial Pearson correlations for the elliptical subsample. Raw $r = 0.07–0.09$; after controlling for log(MassA) all partial correlations collapse to $r \approx 0$ ($p > 0.3$). At the group level (N = 242 groups) raw r rises to 0.13–0.15 but again collapses after halo control.

Table 1. Partial correlation — ellipticals, controlling for log(MassA)

Environment measure	Raw r	Raw p	Partial r	Partial p	Sig?
Individual level (N = 1,270)					
log(CountInCyl)	+0.066	0.019	-0.029	0.296	No
log(DistanceTo5nn)	-0.090	0.001	-0.006	0.824	No
log(SurfaceDensity)	+0.089	0.002	+0.005	0.869	No
log(AGEDenPar)	+0.071	0.012	+0.023	0.424	No
Group level (N = 242 groups)					
log(CountInCyl)	+0.133	0.039	+0.025	0.700	No
log(DistanceTo5nn)	-0.147	0.022	-0.046	0.480	No
log(SurfaceDen	+0.147	0.023	+0.044	0.492	No

sity)					
log(AGEDenPar)	+0.037	0.572	-0.016	0.808	No

All partial correlations statistically indistinguishable from zero ($p > 0.3$).

4.2 Independent Environment Signal for the Full Population

For all galaxy types (N = 22,531 individuals; 5,775 groups), raw r = 0.18–0.25, remaining significant after halo control (partial r = 0.10–0.16, all $p < 10^{-55}$). Approximately 45–58% of the raw signal survives, indicating a secondary pathway.

Table 2. Partial correlation — full population, controlling for log(MassA)

Environment measure	Raw r	Raw p	Partial r	Partial p	% ret.
Individual level (N = 22,531)					
log(CountInCyl)	+0.240	8×10^{-292}	+0.109	7×10^{-61}	45.6%
log(DistanceTo5nn)	-0.245	3×10^{-304}	-0.124	4×10^{-78}	50.8%
log(SurfaceDensity)	+0.245	1×10^{-304}	+0.124	3×10^{-78}	50.8%
log(AGEDenPar)	+0.178	2×10^{-160}	+0.104	2×10^{-55}	58.4%
Group level (N = 5,775) — mean u−r					
log(CountInCyl)	+0.234	$<10^{-300}$	+0.160	$<10^{-300}$	68.4%
log(DistanceTo5nn)	-0.228	$<10^{-300}$	-0.153	$<10^{-300}$	67.1%
log(SurfaceDensity)	+0.228	$<10^{-300}$	+0.153	$<10^{-300}$	67.1%
log(AGEDenPar)	+0.154	$<10^{-300}$	+0.128	$<10^{-300}$	83.1%

All partial correlations remain highly significant ($p << 0.05$).

4.3 Colour Distribution Shape Evolves Monotonically with Redshift

Table 3 and Figure 1 show Box–Cox λ and skewness against redshift with bootstrap null envelopes. Both parameters lie outside the null at every bin. Skewness shifts from +0.90 at z = 0.026 to −0.82 at z = 0.325 (Spearman r = −1.000, $p = 4.0\times10^{-4}$). Box–Cox λ increases from −0.20 to +2.70 (Spearman r = +1.000, $p = 4.0\times10^{-4}$). The trends are statistically significant features of the GAMA flux-limited sample. At low z the blue cloud dominates the observed population, producing a positively skewed, sub-log-Normal distribution. At high z Malmquist selection progressively removes low-luminosity blue galaxies, leaving a population dominated by the red sequence; the distribution becomes negatively skewed with $\lambda >> 1$. Whether this reflects genuine evolutionary change, selection, or a combination requires a stellar-mass-complete subsample.

Table 3. u−r distribution shape parameters by redshift bin

z centre	N	Box-Cox λ	Skewness	Excess kurtosis
0.026	3,932	-0.200	+0.899	+0.824

0.075	16,579	+0.431	+0.263	-0.750
0.125	29,589	+1.125	-0.148	-1.046
0.175	32,999	+1.694	-0.387	-0.897
0.225	27,116	+1.995	-0.523	-0.623
0.275	33,685	+2.532	-0.745	-0.188
0.325	21,588	+2.701	-0.821	+0.069

Spearman $r(\lambda,z) = +1.000$, $p = 4.0\times10^{-4}$; Spearman $r(Skew,z) = -1.000$, $p = 4.0\times10^{-4}$. All bins lie outside the bootstrap 95% null envelope.

Figure 1. Box–Cox λ (left) and skewness (right) of the u−r colour distribution vs redshift. Grey band = 95% bootstrap null envelope (500 iterations). Red stars mark bins where the observed value lies outside the null. Both parameters show perfect monotonic trends (Spearman $|r| = 1.00$, $p = 4\times10^{-4}$).

4.4 λ and Red Fraction as Complementary Distributional Statistics

Table 4 and Figure 2 compare Box–Cox λ with the red fraction f_{red} across the seven redshift bins. Spearman $r(f_{red}, \lambda) = 1.000$: the two statistics are in perfect rank agreement across the flux-limited sample, confirming they respond to the same underlying shift in the observed colour distribution.

Table 4. Red fraction and Box–Cox λ by redshift bin

z centre	N	Red fraction	Box-Cox λ	$\lvert\lambda - 1\rvert$
0.026	3,932	0.142	-0.200	1.200
0.075	16,579	0.219	+0.431	0.569
0.125	29,589	0.336	+1.125	0.125
0.175	32,999	0.405	+1.694	0.694
0.225	27,116	0.434	+1.995	0.995
0.275	33,685	0.468	+2.532	1.532
0.325	21,588	0.477	+2.701	1.701

Spearman $r(f_{red}, \lambda) = +1.000$. Red fraction = fraction of galaxies with $u-r > 2.1$ per bin. $|\lambda - 1|$ measures distance from Normality ($\lambda = 1$ is the Normal point).

Despite this agreement in rank, the two statistics change at different rates as a function of redshift (Figure 2). When both are normalised to [0,1], f_{red} rises more steeply at low-to-mid redshift ($z < 0.2$) while λ continues to increase steeply at $z > 0.2$ where f_{red} growth has largely decelerated ($\Delta f_{red} = 0.009$ per bin at $z = 0.30$–0.35 vs $\Delta\lambda = 0.17$ per bin). The $d\lambda/d(f_{red})$ ratio increases monotonically from ~8 at $z \approx 0.10$ to ~19 at $z \approx 0.30$.

These differing rates reflect the different mathematical sensitivity of each statistic, not necessarily different physical processes. A threshold count f_{red} saturates once most galaxies in the visible sample are already above the threshold; λ, which depends on the full distributional shape, continues to respond to changes in the breadth and asymmetry of the colour distribution even in a predominantly red sample. This saturation behaviour is expected for any threshold statistic and is accentuated in a Malmquist-selected sample where the blue population is progressively lost at high z. We therefore caution against interpreting the diverging rates as evidence of two decoupled physical processes without first establishing results in a mass-complete subsample.

Figure 2. Left: Box–Cox λ and red fraction ($u-r > 2.1$) vs redshift on dual y-axes. Centre: both statistics normalised to [0,1] — shaded gap shows where f_{red} leads (pink) or λ leads (blue). Right: rates of change per redshift bin, illustrating the different mathematical saturation behaviour of a threshold count vs a full distributional statistic in a flux-limited sample.

5. Discussion

Our partial correlation results align with the picture that halo mass is the primary driver of colour–environment correlations for morphologically selected red galaxies (Peng et al. 2012; Woo et al. 2013). Complete mediation for ellipticals suggests that once a galaxy has settled into a massive halo and undergone morphological transformation, residual local environment adds little further quenching pressure. The ~50% survival of the signal in the full population points to secondary processes consistent with harassment and tidal stripping acting preferentially on late-type discs (Moore et al. 1996).

The distribution shape results demonstrate that both Box–Cox λ and skewness are statistically significant features of the GAMA colour distribution at each redshift bin, consistently lying outside the bootstrap null envelope. The perfect rank agreement between λ and f_red (Spearman $r = 1.000$) confirms they are tracking the same fundamental property of the observed sample: the progressive shift from a blue-cloud-dominated to a red-sequence-dominated colour distribution across the redshift range.

However, because GAMA is flux-limited at $r < 19.8$, the observed trends with redshift in any per-bin statistic reflect a combination of genuine galaxy evolution and Malmquist selection — at higher z, only the most luminous, typically red galaxies remain above the detection threshold. The rates of change of λ and f_red with redshift are therefore properties of the flux-limited observed sample and cannot be directly interpreted as evolutionary rates. The diverging rates of the two statistics at $z > 0.2$ are consistent with the known saturation behaviour of threshold statistics when the sample becomes dominated by a single population, amplified by the progressive loss of blue galaxies from the flux-limited sample. Establishing whether λ provides additional discriminating power for physical processes requires stellar-mass-complete volume-limited subsamples — for example, using the GAMA stellar mass completeness estimates or extending the analysis to deeper photometric surveys such as DES or LSST, where comparable mass thresholds can be maintained to $z \sim 1$.

Notwithstanding the selection caveat, the comparison of λ and f_red demonstrates a practical statistical point: a Box–Cox power transform captures the shape of the full colour distribution and does not saturate as a threshold count does. In a regime where f_red is near its ceiling, λ retains sensitivity to structural changes in the distribution. This makes λ a potentially useful complement to red fraction in studies where the galaxy population is predominantly red.

6. Conclusions

1. Halo mass fully mediates colour–environment for elliptical galaxies (N = 1,270); partial correlations collapse to $r \approx 0$ after controlling for log(MassA).
2. For the full galaxy population (N = 22,531), ~50% of the signal survives halo control, indicating a morphology-linked secondary pathway.
3. Skewness and Box–Cox λ evolve monotonically with redshift in the GAMA flux-limited sample (|Spearman r| = 1.00, $p = 4\times10^{-4}$), consistently lying outside the bootstrap null envelope. These trends reflect a real shift in the observed colour distribution, though their interpretation as evolutionary rates is limited by Malmquist selection; a mass-complete subsample is required to separate selection from evolution.
4. Box–Cox λ and the red fraction are in perfect rank agreement (Spearman $r = 1.000$) but diverge in rate at $z > 0.2$, consistent with the mathematical saturation of a threshold statistic in a flux-limited, increasingly red sample. λ retains sensitivity to distributional shape changes where f_red has plateaued, suggesting it may complement threshold-based statistics in predominantly red samples.

5. Extension to mass-complete samples and deeper photometric surveys (DES, LSST) is needed to determine whether the shape evolution reflects genuine quenching history.

Data Availability

GAMA DR4: https://www.gama-survey.org/dr4/. Analysis notebooks available on request.

Appendix A: Johnson SU Fit Parameters

Table A1 gives the full set of shape parameters including Johnson SU (a, b). The JSU a parameter shows large non-monotonic variation despite the smooth evolution of skewness and λ, arising from correlation between a and b on a near-flat likelihood ridge.

Table A1. Full shape parameters including Johnson SU

z	N	JSU a	JSU b	Box-Cox λ	Skewness	Kurt
0.026	3,932	-2.763	+2.210	-0.200	+0.899	+0.824
0.075	16,579	-11.677	+5.925	+0.431	+0.263	-0.750
0.125	29,589	+20.679	+7.171	+1.125	-0.148	-1.046
0.175	32,999	+9.776	+3.566	+1.694	-0.387	-0.897
0.225	27,116	+3.431	+2.112	+1.995	-0.523	-0.623
0.275	33,685	+2.802	+1.752	+2.532	-0.745	-0.188
0.325	21,588	+2.836	+1.802	+2.701	-0.821	+0.069

JSU a: Spearman $r = +0.29$, $p = 0.53$ (not significant). Box–Cox λ and skewness: $|r| = 1.00$, $p < 0.001$. JSU a varies by >30× while skewness changes by only 0.75 — instability is numerical, not physical.

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